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Enhancing Weather-Related Power Outage Prediction by Event Severity Classification

FEIFEI YANG¹, PETER WATSON, MARIKA KOUKOULA, AND EMMANOUIL N. ANAGNOSTOU

Department of Civil and Environmental Engineering, University of Connecticut, Storrs, CT 06269, USA

Corresponding author: Emmanouil N. Anagnostou (manos@uconn.edu)

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ABSTRACT The accuracy of machine learning–based power outage prediction models (OPMs) is sensitive to how well event severity is represented in their training datasets. Unbalanced or overly dispersed event severity can result in random errors in outage predictions and underestimation in severe events or overestimation in weak ones. To improve accuracy in the prediction of storm-caused power outages, we introduce a novel method called “Conditioned OPM” that divides an OPM training dataset into subsets of events representative of the predicted event’s severity by calculating the quantile weight distance (QWD) between severe weather-related events in the dataset and the predicted event. Based on 102 storm events (including two hurricanes, Irene and Sandy), that have occurred since 2005 over Eversource Energy’s Connecticut service territory, we quantified the weather differences among predicted events, which we classified into three groups of severity: low, moderate, and high. The Conditioned OPM creates a subset of the historical events based on their classified severity group and uses that subset as the training dataset to predict the power outages. The study shows that the accuracy of event severity classification was 0.76, and the mean absolute percentage error (MAPE) decreased by about 30%; this method was also tested on forecast events and exhibited a low (20%) MAPE.

INDEX TERMS Event severity classification, severe weather, machine learning, quantile weight distance, outage prediction model, model training.

I. INTRODUCTION

Threats to the security of power systems are classified into four categories: natural, accidental, malicious, and emerging [1]. Severe weather is one natural threat that can cause many power outages, costing the national economy billions of dollars [2] and affecting millions of utility customers in the United States each year [3]. The accurate prediction of power outages before severe weather events take place could reduce these costs by enabling utility companies to improve their preparation of equipment and their crew allocation to restore power [4].

The literature describes a best practice of using machine learning (ML) techniques in the electric power system and using ML to predict power outages [5]–[18]. Nateghi *et al.* [19] used random forest (RF) to predict outages caused by tropical cyclones, and Wanik *et al.* [20] used tree-based models to predict power outages in New England. He *et al.* [21] and Cerrai *et al.* [22], [23] utilized Bayesian

additive regression trees (BART) to predict outages from storm events, while Eskandarpour and Khodaei [24] explored artificial neural network logistic regression to predict outages from an imminent hurricane. Kabir *et al.* [25] utilized RF, boosting tree (BT), support vector machine (SVM), and quantile regression forest (QRF) for predicting thunderstorm-induced power outages. These multiple ML approaches have pushed forward the ability to predict power outages for electric distribution networks.

Although outage prediction modeling has thus achieved improvements through ML methods, current models [19]–[25] often use the same input training datasets of historical events for predicting events at differing levels of severity. Some previous studies have taken the approach of using diversity sampling with balanced class distribution in their training datasets [26]–[29], few investigate predicting power outages by dividing the training dataset into subsets of events representative of the tested event’s severity. The sensitivity of ML models to their training datasets [30] results in biases and random errors in the predictions, which limits their accuracy. Yang *et al.* [31] have demonstrated that

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predicting power outages by dividing the training dataset into subsets of events representative of the tested event's severity could significantly reduce the bias. In Yang *et al.*'s previous research actual outages information of the severe event was used to categorize that event and selected historical events in that category were used to train the OPM. The limitation of this approach is that the method cannot apply for forecasting since the actual outages of an upcoming event are unknown.

Some studies have taken a similar approach to this one. Shashaani *et al.* [32], for example, used a binary classification model to predict whether a grid cell would have power outages, and then developed a power outage prediction model for those that would. The study used only five hurricanes in its model, however. Loewe and Prölss [33] classified geomagnetic storms as weak, moderate, strong, severe, and great using the minimum disturbance storm time index—a description of the severity of geomagnetic events—but this classification was not especially related to power networks.

In this study, we extend Yang *et al.*'s research on OPM training data representativeness. We sought to gain a better understanding of the threat posed to the power system by storms by grouping them in terms of severity, that is, based on the number of outages the storms caused in Connecticut. Specifically, storms were grouped by outage ranges corresponding to thresholds used by Connecticut utility company for pre-storm allocation of restoration crews: low severity (less than 100 outages), moderate severity (100 to 1,000 outages), and high severity (more than 1,000 outages). To classify an upcoming event to the severity levels defined above, we applied a novel statistical method that quantifies the differences among historical events in the quantile weight distance (QWD) of weather variables. Next, we selected the training dataset of each event by the level at which it was classified according to the QWD-based method. As the event severity classification depends on forecasted weather variables, and not on the number of actual outages, this method could be applied to outage forecasting.

Our study has three main objectives. First, we set out to develop the methodology for classifying events into low-, moderate-, and high-severity groups by calculating the QWDs of key weather variable CDFs (cumulative density functions) of the events. Second, we demonstrated through a storm-based holdout test that the event severity classification improved outage prediction model (OPM) performance by dividing the training dataset into subsets and selecting optimization coefficients for the predicted events. Third, we demonstrated that the severity classification of forecasted events would significantly reduce the bias of the real-time OPM-based forecasting system.

II. STUDY AREA AND DATA

Our study focused on the Connecticut service territory of Eversource Energy, which covers 149 towns across the state. We investigated outage events of one hundred extratropical storms and two hurricanes that took place in Connecticut between 2005 and 2017. Extratropical storms last from

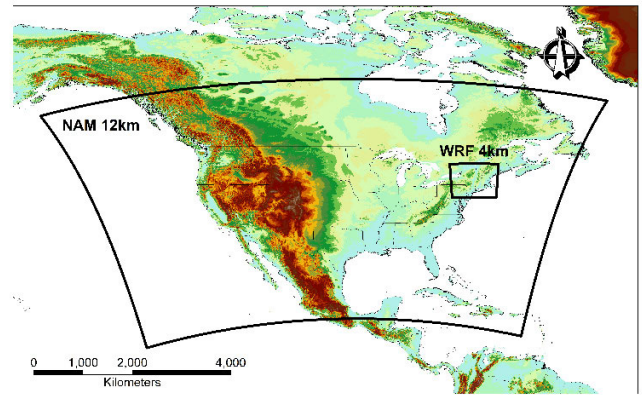


FIGURE 1. NAM and WRF domains.

several hours to several days and exhibit strong sustained winds and/or long periods of rain. Hurricanes are among the most prominent natural hazards, often exhibiting very high winds and rains. Both kinds of storms cause many power outages in the service territory. Here we integrated weather variables, utility infrastructure, land cover, vegetation, and historical power outages for each storm event and modeled the power outages to a 4-km resolution with 803 grid cells covering the region. Details of the data sources follow.

A. WEATHER

Weather input is the largest source of uncertainty in the OPM predictions [34], [35]. To improve their accuracy, we derived the weather data used in the OPM in this study from 48-hour simulations using version 3.8.1 of the Advanced Research Weather Research and Forecasting (WRF) model [36], [37], at a 4-km horizontal grid spacing over the northeastern United States (Figure 1). For initial and lateral conditions in our simulations, we used weather forecasts and analysis from the North American Mesoscale (NAM) Forecast System at 12-km spatial resolution. Specifically, we used analysis data to investigate the QWD event classification and the Conditioned OPM, and then we apply the QWD event classification and the Conditioned OPM to the forecast system by using forecast data. Figure 1 shows the NAM and WRF simulation domains. The weather variables used in the current OPM, derived for each event from a 48-hour time series of WRF outputs for each grid point, are shown in Table 1 and further explained in Section III. The “MAX” variables (such as “MAXGust”) are the 48-hour maximum values of each variable, and the “MEAN” variables (such as “MEANTemp”) represent the mean values during a 4-hour window of the strongest winds.

B. UTILITY INFRASTRUCTURE

The utility infrastructure in the Connecticut service territory of Eversource Energy contains multiple types of assets, including electric fuses, reclosers, and poles. We geographically aggregated utility infrastructure data to the WRF model's inner domain 4-km grid cells and provided

TABLE 1. Explanatory variables included in the OPM.

Variables	Abbreviation	Description	Units
Duration of wind at 10 m above 5 m/s	wgt5	Weather	hr
Duration of wind at 10 m above 9 m/s	wgt9	Weather	hr
Duration of wind at 10 m above 13 m/s	wgt13	Weather	hr
Duration of wind at 10 m above 18 m/s	wgt18	Weather	hr
Continuous hours of wind above 5 m/s	Cowgt5	Weather	hr
Continuous hours of wind above 9 m/s	Cowgt9	Weather	hr
Continuous hours of wind above 13 m/s	Cowgt13	Weather	hr
Continuous hours of wind above 18 m/s	Cowgt18	Weather	hr
The last hour when wind above 9 m/s	mxhw9	Weather	hr
Maximum amount of water contained in the foliage canopy	MAXCANWAT	Weather	kg/m ²
Maximum value of low-level turbulence that occurs below 1500 ft MSL	MAXLLTURB	Weather	-
Maximum wind speed at 10 m height	MAXWind10m	Weather	m/s
Maximum wind gust	MAXGust	Weather	m/s
Maximum mass of water vapor in a unit mass of moist air	MAXSpecHum	Weather	kg/kg
Maximum precipitation rate	MAXPreRate	Weather	mm/hr
Maximum soil moisture	MAXSoilMst	Weather	mm/mm
Maximum surface runoff	MAXSfcRoff	Weather	kg/(m ² s)
Maximum value of low-level turbulence that occurs below 1500 ft MSL	MEANLLTURB	Weather	-
Mean wind at 10 m height	MEANWind10m	Weather	m/s
Mean wind gust	MEANGust	Weather	m/s
Mean mass of water vapor in a unit mass of moist air	MEANSpecHum	Weather	kg/kg
Mean temperature	MEANTemp	Weather	K
Total accumulated precipitation	MaxTotPrec	Weather	mm
Percent of miscellaneous forest	percXFrst	Land cover	%
Percent of deciduous forest	percDecid	Land cover	%
Percent of developed area	percDevel	Land cover	%
The count of electric poles	cntPole	Infrastructure	count
The count of fuses and reclosers	sumIso	Infrastructure	count
Leaf area index (leaf area/ground area)	LAI	Vegetation	m ² / m ²

“cntPole”—that is, a count of poles per 4-km grid cell—and “sumIso”—the sum of fuses and reclosers per 4-km grid cell—for the OPM. Shown in Table 1, these are the most important explanatory input variables in a trained OPM. This is because outages are recorded at the asset level, and the risk of having a reported outage is directly proportional to the number of assets.

C. LAND COVER

National Land Cover Database (NLCD) products with detailed vegetation and urbanization patterns were provided by the U.S. Geological Survey (USGS) [38]. Previous studies [20], [22] have shown the method of generating land cover variables used in the OPM. The interaction of trees next to the overhead lines with the infrastructure during storms

is the major cause of outages, so we used in this study tree-related land cover variables (that is, the percentages of miscellaneous forest, deciduous forest, and developed area), which we aggregated into 4-km grid cells. The land cover variables are shown in Table 1.

D. VEGETATION

The seasonal variability of the number of leaves on trees is not explained by land cover variables, so we used the weekly climatological leaf area index (LAI, describes the amount of foliage in the plant) to indicate it. LAI is based on observations from the Moderate Resolution Imaging Spectroradiometer (MODIS) aboard NASA's Terra and Aqua satellites [39], processed to create a weekly climatological product [22]. The LAI was sampled at the WRF 4-km resolution grid.

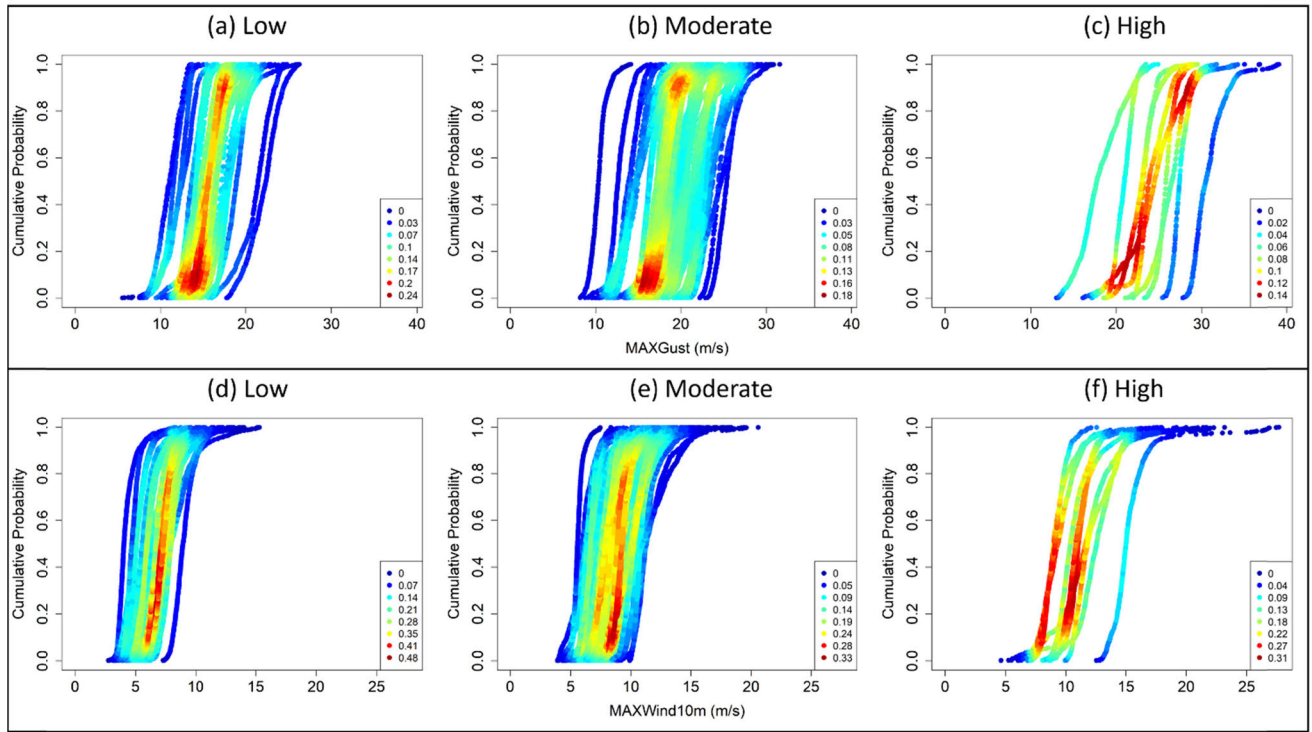


FIGURE 2. CDF density plots for MAXGust (a, b, c) and MAXWind (d, e, f) for 31 low-severity events (a, d), 61 moderate-severity events (b, e), and 10 high-severity events (c, f).

E. HISTORICAL POWER OUTAGES

Historical power outages were reported, with starting times and geographical coordinates, by the utility's outage management system (OMS). The outages were aggregated and counted during each storm period per 4-km grid cell and the numbers then utilized in the OPM.

III. EVENT SEVERITY CLASSIFICATION METHODS

We divided the 102 events in Connecticut into three severity levels: low (causing less than 100 outages), moderate (100 to 1,000 outages), and high (above 1,000 outages). It is reasonable to classify the severity of events based on their weather variables, since the number of outages is a function of the severity of the weather. Therefore, the difference in weather parameters could represent the difference among severe weather events. We developed a quantile weight distance (QWD) method to measure the relatedness of weather variable CDFs of storm events and, thus, to classify the storm events into the low-, moderate-, and high-severity groups. Figure 2 shows the CDF density plots of MAXGust (panels a, b, and c) and MAXWind10m (panels d, e, and f), which are two crucial weather parameters used in the OPM. The figure shows that, for both MAXGust and MAXWind10m, the distributions of the same variables varied across event severity (the density increases following blue-yellow-red in Figure 2).

The QWD of two discrete distributions, sample A ($x_1, x_2, \dots, x_N$ in ascending order) and sample B

($y_1, y_2, \dots, y_N$ in ascending order), is calculated as follows:

$$\text{QWD} = \frac{\sum_{i=1}^N (X_i - Y_i) F_i^k}{\sum_{i=1}^N F_i^k}, \quad (1)$$

where F_i is the cumulative probability distribution ($1/N, 2/N, 3/N, \dots, 1$) at step number i ; x_i and y_i are original weather variables; X_i and Y_i are the weather parameter values at F_i , scaled by the number of utility infrastructure assets in the corresponding x_i and y_i grid cells; k is the root number, with higher k making the difference stronger, and here, $k = 3$. The scaling of weather variable values by the number of infrastructure assets was introduced to account for the spatial colocation of weather and electric infrastructure, which impacts the outage severity of an event. N is the number of grid cells, and here, $N = 803$. i is the step number, which ranges from 1 to N . QWD carries the units of the weather parameter, where $\text{QWD} > 0$ means the values of weather parameters of sample A are higher than those of B; $\text{QWD} = 0$ means the difference between A and B can be ignored; and $\text{QWD} < 0$, means the values of sample A are less than those of B.

For different weather variables, the QWD should have different units and according to the formula (1). However, we will only compare the QWD values between the same variable, and the grouping will be based on threshold values that are unique to each variable (this is described below).

Since MAXGust is the key weather variable used by utilities to predict the impact of a severe event, we have chosen it to demonstrate how we calculated the QWD and

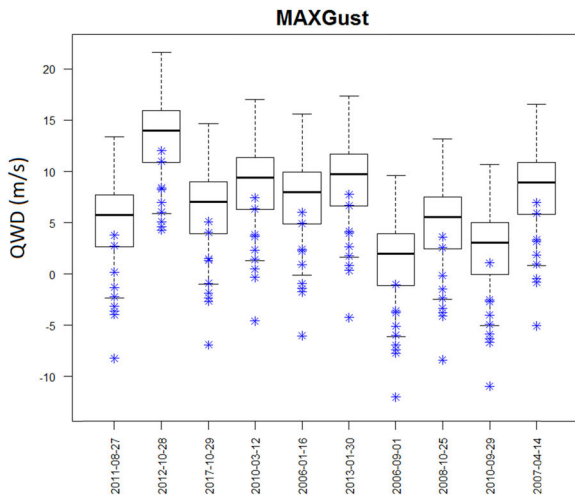


FIGURE 3. QWD of CDFs between each of 10 high-severity events and each of the remaining 92 events (boxplots), as well as the QWD of CDFs among 10 high-severity events (blue star points) for not normalized MAXGust to judge if MAXGust is a significant weather variable.

determined the significant weather variables used in the OPM. Figure 3 shows the QWD results of MAXGust CDFs between each of the 10 high-severity events and each of the remaining 92 events (boxplots shown in the figure), as well as the QWD results of MAXGust CDFs among the 10 high-severity events (blue star points). The weather data used in the calculation of QWD (Figure 3) are not normalized by the utility infrastructure, which means we used original weather variables x_i and y_i to calculate QWD (e.g. change X_i to x_i , change Y_i to y_i in equation (1)). This is because for this step we need to identify the weather variable which differ between the events with different severity. This should not be related to infrastructure asset distribution. From Figure 3, it shows the difference of QWD values of MAXGust between the high-severity events and moderate/low-severity events. After calculating the QWD values, we used t-tests to determine whether the means of the boxplots and of the blue star points differed significantly in Figure 3. We referred to as statistically significant as $P < 0.05$, and, finally, found the significant weather variables.

To classify event severity by QWD methods, we developed a four-step heuristic, depicted in Figure 4.

For the first step, we calculated the QWD of CDFs between each of the 10 high-severity events (X in equation (1)) and each of the remaining 92 events (Y in equation (1)), as well as the QWD of CDFs between each of the 10 high-severity events (X in equation (1)) and each of the remaining 9 high-severity events (Y in equation (1)) for the 23 significant weather variables described in Table 1.

We calculated the average percentage (y-axis in Figure 5) of the blue star points of these 10 high-severity events below the 0%, 5%, 10%, 15%, 20%, and 25% quantiles (x-axis in Figure 5) of their corresponding boxplots. From Figure 5, we picked the six key variables (Cowgt13, Cowgt18, MAXLLTURB, MEANLLTURB, MAXGust, and MEANGust) that exhibited the highest average percentages

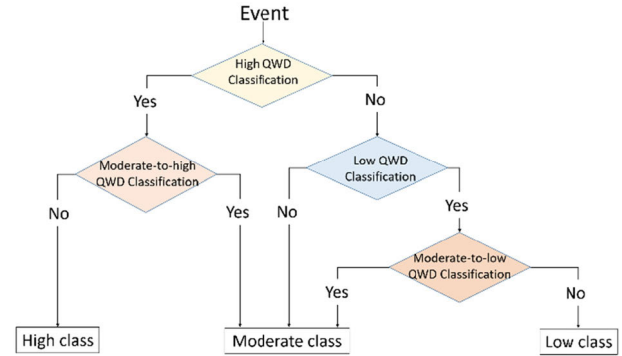


FIGURE 4. Decision tree of event severity classification by QWD methods.

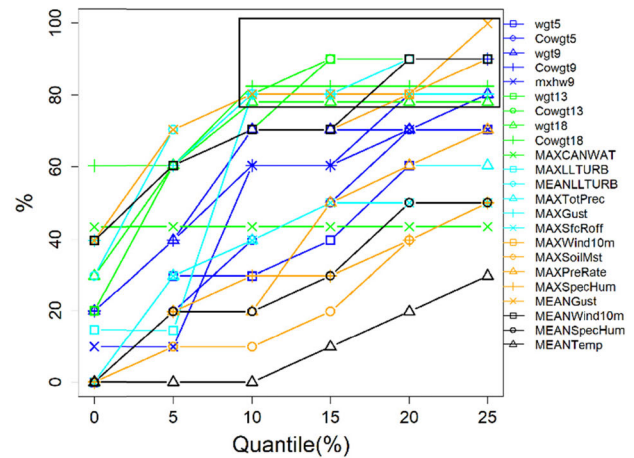


FIGURE 5. Average percentage (y-axis) of blue star points of these 10 high-severity events below the 0%, 5%, 10%, 15%, 20%, and 25% quantile (x-axis) of their corresponding boxplots in the step one decision.

(above 80) of blue star points of these 10 high-severity events below the 10% quantile of their corresponding boxplots (six marked variables in the black rectangle area), and we set the QWD values of the 10% quantile of the boxplots as the threshold of step one decision. Thus, the threshold of step one decision exhibited 10 QWD values for each key variable (that is, each of 10 high-severity events exhibited one QWD value as the threshold at the 10% quantile of the boxplots).

For the calculation by equation (1), X was the high-severity event, and Y was the predicted event. If the QWD value between the high-severity event and the predicted event was at or below the threshold value, it meant the QWD value was small (that is, the weather difference between the high-severity event and the predicted event was small), and Y was close to the high-severity event; thus, the predicted event was classified as a high-severity event (Yes). Otherwise, if the QWD value between the high-severity event and the predicted event was above the threshold, then this event was classified as a not-high severity event (No). For each of the 10 high-severity events, there were 54 prediction results (that is, there were 54 QWD results between 9 high-severity events and the predicted high-severity event for the six key variables). For total 10 high-severity events, it was 540 prediction results (“YES” or “NO”) for these six variables.

For each of the 92 moderate- and low-severity events, there were 60 prediction results (that is, 60 QWD results between each of 10 high-severity events and the predicted event for the six key variables). We combined all the classification results from these six variables and found another probability threshold to make the final decision, here “0.74” is this threshold. Such that if the calculated probability for any event of being “high-severity” was at or above 0.74, it was classified as a high-severity event; otherwise, it was classified as a not-high severity event. The probability value of “0.74” was obtained by evaluating the probability values from 0.5 to 1 as thresholds to see the impact on prediction accuracy for all the events, and “0.74” exhibited the highest prediction accuracy for both 10 high-severity events and remaining 92 not-high severity events. As noted, this process is called “High” QWD classification.

In steps two, three, and four, we used similar processes to classify event severity, as described in step one. The steps are summarized as,

Step one: Classify the event as a high-severity event (Yes) or not (No); this is called high QWD classification.

Step two: Classify an event determined as Yes in the first step as a moderate-severity event (Yes) or not (No); this is called moderate-to-high QWD classification. If No, then the event is classified as high. If Yes, then the event is classified as moderate.

Step three: Classify an event determined as No in the first step as a moderate-severity event (No) or not (Yes). If Yes, then go to step four; this is called low QWD classification.

Step four: Classify an event determined as Yes in the third step as a moderate-severity event (Yes) or not (No); this is called moderate-to-low QWD classification. If NO, then the event is classified as low.

Using these high, low, moderate-to-high, and moderate-to-low QWD classifications, we classified the severity level of the 102 events included in the study.

The differences among the four QWD classification steps—high, low, moderate-to-high, and moderate-to-low—are summarized in Table 2. The figures used for selecting the key variables for steps two, three, and four are shown in the appendix.

We used the confusion matrix [40], [41] to show the performance of the classification results for the 102 events. Specifically, we calculated classification accuracy for three class labels using the following metrics: TP (true positive); FN (false negative), Precision; Recall [42], [43]; and F1 Score [44] for each class label. The explanations of TP and FN, and the formulas for Accuracy, Precision, Recall, and F1 Score are described in the appendix.

IV. OUTAGE PREDICTION MODEL

The OPM presented in this study used multiple tree-based ML regression algorithms (BART, RF, and ensemble decision tree (ENS)), which had been used in previous studies [22], [23].

The ENS uses multiple learning algorithms, which may yield better prediction performance. In our study, the ENS model combined DT, RF, and BT models and fitted recursive partitioning (RPART) to create a decision tree from the outputs of these combined models. The estimation from the ENS model was used for predicting the power outages in the service territory. These BART, RF, and DT models are described in detail in the appendix.

We structured the OPM using weather analysis (for training) or forecast (for prediction). We used weather, utility infrastructure, land cover, and LAI data as the input variables, called predictors X , which included all information related to power outages, and used actual power outages as the target variable Y . We split the historical event data into training and testing datasets, both containing predictors X and target variable Y . We created ML models relating to X and corresponding Y from the training data and used them to predict target variable Y of the testing data. For different ML regressions, the created models and outputs were different—that is, predicted Y s were different for BART, RF, and ENS with the same predictors X input of the testing data. The outage prediction output for each event was given by a linear combination of predictions from the BART, RF, and ENS models, based on equation (2):

$$P_j = CM_j = \sum_{i=1}^3 c_i m_{ij}, \quad (2)$$

where P_j is the power outage prediction for the j -th storm event; $M_j = [m_{1j} \ m_{2j} \ m_{3j}]^T$ and m_{1j} , m_{2j} , m_{3j} represent outage prediction outputs from the BART, RF, and ENS models for the j -th storm event, and $C = [c_1 \ c_2 \ c_3]$, and c_1 , c_2 , c_3 represent the weights of BART, RF, and ENS predictions, respectively.

The coefficients of C were calculated by optimizing the model performances according to the least square difference of predictions P to actual O equation (3):

$$\argmin \|P - O\|_2 \quad \text{subject to } C \geq 0, \quad (3)$$

where O represents actual outages for predicted events, and coefficients C are restricted by nonnegativity, as the outages value is nonnegative. In this study, the optimization coefficients C varied across the event severity.

V. CONDITIONED OPM

As mentioned in sections I and III, the ML models are sensitive to how different event severities are represented in the training dataset. Selecting different training datasets for events associated with different severity levels could reduce the bias in the OPM. In this study, we defined a new model called the Conditioned OPM that used the QWD-based severity classification described in section III to divide the training dataset into subsets of events representative of the severity of the predicted event to predict the power outages caused by each of the 100 extratropical events. The training dataset we used comprised all 100 extratropical events, and each test case

TABLE 2. Significant differences in four-step decisions.

Step	QWD between different severity events	QWD between same severity events	Key weather variables	Threshold Setting
<i>Step one decision: “High” QWD classification.</i> The predicted event was classified as a high-severity event when the calculated probability for this event of being “high-severity” was at or above the probability value of step one decision. Otherwise, it was classified as a not-high severity event.	QWD of CDFs between each of the 10 high-severity events and each of the remaining 92 events	QWD of CDFs among the 10 high-severity events	Cowgt18, Cowgt13, MAXLLTURB, MEANLLTURB, MAXGust, MEANGust	Set the QWD value at the 10% quantile of the boxplots as the threshold of the decision
<i>Step two decision: “Moderate-to-high” QWD classification.</i> The predicted event was classified as a moderate-severity event when the calculated probability for this event of being “moderate-severity” was at or above the probability value of step two decision. Otherwise, it was classified as a high-severity event.	QWD of CDFs between each of the 61 moderate-severity events and each of the 10 high-severity events	QWD of CDFs among the 61 moderate-severity events	MEANGust, wgt13, wgt18, MAXLLTURB, MAXGust, MAXWind10m	Set the QWD value at the 65% quantile of the boxplots as the threshold of the decision
<i>Step three decision: “Low” QWD classification.</i> The predicted event was classified as a low-severity event when the calculated probability for this event of being “low-severity” was at or above the probability value of step three decision. Otherwise, it was classified as a moderate-severity event.	QWD of CDFs between each of the 31 low-severity events and each of the remaining 71 severity events	QWD of CDFs among the 31 low-severity events	MAXWind10m, MEANWind10m, wgt9, Cowgt9, MEANGust	Set the QWD value at the 65% quantile of the boxplots as the threshold of the decision
<i>Step four decision: “Moderate-to-low” QWD classification.</i> The predicted event was classified as a moderate-severity event when the calculated probability for this event of being “moderate-severity” was at or above the probability value of step four decision. Otherwise, it was classified as a low-severity event.	QWD of CDFs between each of the 61 moderate-severity events and each of the 31 low-severity events	QWD of CDFs among the 61 moderate-severity events	MEANWind10m, MAXWind10m, mxhw9, MAXLLTURB, MEANGust, wgt9	Set the QWD value of the 10% quantile of the boxplots as the threshold of the decision

was held out of the training data of the Conditioned OPM. A point to note is that the two hurricanes in our dataset were used only for QWD classification of the extratropical events.

As also mentioned in the previous sections, we grouped the events into three severity levels: low, moderate, and high. Since the training dataset of historical events resulting in fewer than 200 power outages exhibited the best model performance in predicting holdout low-severity events, we chose it as the training dataset for predicting such events in the Conditioned OPM. Since the training dataset including all the historical events exhibited the best model performance in predicting moderate- and high-severity events, we chose it as the training dataset for predicting these events. We calculated

the optimization coefficients C_{low} for predicting events classified as low-severity, $C_{moderate}$ for predicting events classified as moderate-severity, and C_{high} for predicting events classified as high-severity. These optimization coefficients were calculated by optimizing the model performances according to the least square difference of predictions P to actual O of all predicted low-/moderate-/high-severity events, respectively.

To demonstrate the improvement obtained by using the Conditioned OPM, we compared its performance with that of the Standard OPM for both the weather analysis and forecasts. We used leave-one-out cross-validation (LOOCV) in the Standard OPM and the same training dataset of historical events for the validations. The predicted event was

held out from the training dataset in the Standard OPM. The optimization coefficient for Standard OPM was C_{st} , which was also calculated by optimizing the least square difference of predictions P to actual O of all predicted events.

To demonstrate the effectiveness of the QWD event severity classification for forecasting purposes, we compared the performance of the Conditioned OPM using analysis weather data and forecast weather data, respectively, in predicting 14 extratropical events that took place from May to December 2017 across Connecticut. These events represent the most severe cases that affected the Eversource – Connecticut service territory in that period. Among these high-severity event is the October 29, 2017 bomb cycle that caused nearly 5000 outages and was severely underestimated by the standard OPM. These 14 events are listed in the appendix.

We evaluated the outage prediction model performance according to several error metrics, including AE (absolute error), APE (absolute percentage error), MAPE (mean absolute percentage error), CRMSE (centered root mean square error), R^2 (coefficient of determination R-squared), and NSE (Nash-Sutcliffe efficiency), which are described in the appendix. For the error metrics, smaller AE, MAPE, and CRMSE mean better model performance, and larger R^2 and NSE mean more advanced model performance.

VI. RESULTS AND DISCUSSIONS

A. EVENT CLASSIFICATION RESULTS

This section discusses the event severity classification results based on the QWD methods. The confusion matrix of the classification results for the 102 events is shown in Table 3. The true classes we used here were those defined in Section III. Table 3 shows that both hurricanes (which resulted in 16,045 outages on August 27, 2011, and 15,314 outages on October 28, 2012, respectively) were predicted as high-severity events. Among the remaining 8 high-severity extratropical events, 6 were predicted as high-severity events and 2 (resulting in 1,352 outages on September 1, 2006, and 1,141 outages on September 29, 2010) as moderate-severity events. Among the 61 moderate-severity extratropical events, 19 with actual outages below 312 were predicted as low-severity events and one with 823 actual outages (event on October 27, 2006) as a high-severity event. Among the 31 low-severity extratropical events, two (with 71 outages on March 10, 2017, and 60 outages on January 7, 2015, respectively) were predicted as moderate-severity events. The moderate-severity events which were classified as low-severity events were from the lower bound of the range (100 to 1,000 outages), and the moderate-severity event which was classified as the high-severity event was from the upper bound of that range.

Classification accuracy was calculated by equation (4):

$$Accuracy = \frac{29 + 41 + 8}{102} = 0.76$$

TABLE 3. Confusion matrix for event severity classification for 100 extratropical events and two hurricanes.

Predicted \ True	Low	Moderate	High
Low	29	2	0
Moderate	19	41	1
High	0	2	8

TABLE 4. TP, FP, Precision, Recall, and F1 score of event severity classification results for each class label.

Class	TP	FN	Precision	Recall	F1 Score
Low	29	2	0.60	0.94	0.73
Moderate	41	20	0.91	0.67	0.77
High	8	2	0.89	0.80	0.84

Table 4 shows TP, FN, Precision, and Recall of event severity classification results for each class label calculated by equations (5–7).

The Accuracy, Precision, and Recall results and F1 Score close to 1 indicate accurate classification results. The classification accuracy for the 102 storm events was 0.76. Table 4 shows that, for classification metrics of each class label, the label class of moderate-severity events exhibited the best Precision (0.91) relative to the label classes of high- and low-severity events; the label class of low-severity events exhibited the best Recall (0.94) relative to the label classes of high- and moderate-severity events; and the label class of high-severity events exhibited the best F1 Score (0.84) relative to the label classes of moderate- and low-severity events. The combination of confusion matrix, classification accuracy, and metrics classification results showed acceptable classification performance of the QWD method.

Of the 102 events, 41 were associated with 100 to 300 actual outages, accounting for two-thirds of all the defined moderate-severity events. The CDF density plots for MAXGust for the 31 low-severity events (a) and the 61 moderate-severity events (b) exhibited similar distributions, which are shown in Figure 2. We believe the similarity of weather characteristics highlights the fact that the outages are also related to vegetation and land cover information, which were not included in the classification which we will consider in the future research; thus, 19 out of the 61 moderate-severity events with outages below 312 were classified as low-severity events by QWD decision trees, and this is the reason the Precision of low-severity event classification was 0.60 and the Recall of moderate severity event classification was 0.67.

Generally speaking, a utility company will make a scale table to categorize the severity level of a past storm event according to its wind speed, wind gust, and precipitation, since the utility has thresholds of these characteristics for

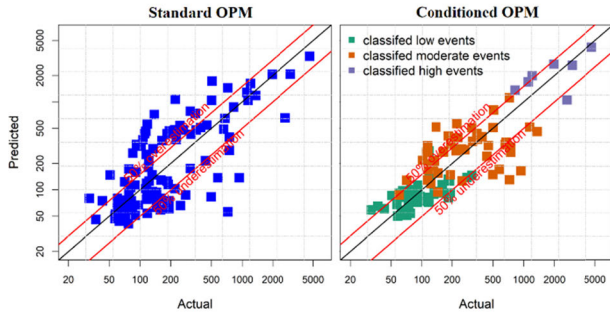


FIGURE 6. Actual outages vs. predicted outages for the Standard OPM and Conditioned OPM for 100 extratropical events.

the different severity levels of such events. Little academic work has been done, however, to classify the outage severity levels of storms based on the forecast weather data. The application of event severity classification to the forecast weather data could improve outage prediction, thus helping utilities understand the severity levels of upcoming storms and improving their ability to manage them.

B. CONDITIONAL OPM PERFORMANCE

This section discusses the outage prediction performances of the Standard OPM and the Conditioned OPM (previously mentioned in Section V) for the 100 extratropical events. Figure 6 shows the validation results of the two models for the 100 storm events. In each panel in the figure, the vertical and horizontal axes represent the log-scale predicted and actual outages of each event, respectively. The parallel red lines represent 50 percent model overestimation (the top red line) and 50 percent model underestimation (the bottom red line), whereas the black line is the 45-degree line at which predicted and actual agree.

Figure 6 shows that the Conditioned OPM predicted more events within the $\pm 50\%$ error bounds and performed better than the Standard OPM, which exhibited more events under- or overestimated by more than 50%. We evaluated the outage prediction performance of the Standard and Conditioned OPMs using several error metrics, including AE, APE, MAPE, CRMSE, and NSE (calculated by equations (9–13)). The evaluation metrics (MAPE, CRMSE, R2, and NSE) of the Standard and Conditioned OPMs are shown in Table 5.

In a comparison of error changes from the Standard OPM to the Conditioned OPM, for the Conditioned OPM, AE Q25, AE Q50, and AE Q75 were lower by 8, 43, and 69 outages, respectively; APE Q25, APE Q50, and APE Q75 were lower by 3%, 6%, and 25%, respectively; MAPE was decreased by 29%; CRMSE was lower by 77 outages; and R-squared and NSE were higher by 0.12.

Figure 7 is a scatter plot comparing the AE of the Standard OPM to that of the Conditioned OPM. It shows that, for most of the events, the AE of the Standard OPM was higher than that of the Conditioned OPM. The orange square points in the figure highlight the AEs of the three most severe events, which resulted in 4,626 outages (TS) on

TABLE 5. Evaluation metrics of validation results for the standard and conditioned OPMs.

Model	AE Q25	AE Q50	AE Q75	APE Q25	APE Q50	APE Q75
Standard OPM	27	94	234	22%	44%	85%
Conditioned OPM	19	51	165	19%	38%	60%

Model	MAPE	CRMSE	R2	NSE
Standard OPM	75%	371	0.67	0.67
Conditioned OPM	46%	294	0.79	0.79

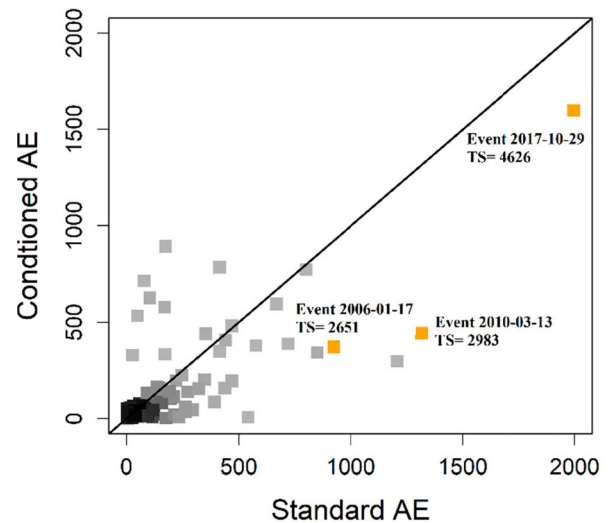


FIGURE 7. AE of the Standard OPM vs. AE of the Conditioned OPM.

October 29, 2017, 2,983 outages on May 13, 2010, and 2,654 outages on January 17, 2006, respectively. The Conditioned OPM exhibited much lower AE than the Standard OPM in all three severe events. This is because the Conditioned OPM considers the representativeness of the training data and uses the optimization coefficients to predict events based on the event severity classification results. This is consistent to finding by Yang *et al.* who showed that conditioning the training of the OPM to a subset of events representative of the predicted event’s severity reduced the prediction bias [31].

In Figure 8, a scatter plot of all the sorted AE (e.g. absolute errors in ascending sequence) of the Conditioned and Standard OPM shows that 92% AE of the Standard OPM is larger than that of the Conditioned OPM. This indicates the Conditioned OPM generates predictions of power outages much closer to the actual outages than those produced by the Standard OPM, with the error significantly decreased.

In sum, the above analysis demonstrates that use of event severity classification and the Conditioned OPM produces better prediction results and lower bias than use of the Standard OPM. The selection of a representative training dataset for each class of predicted events for the Conditioned

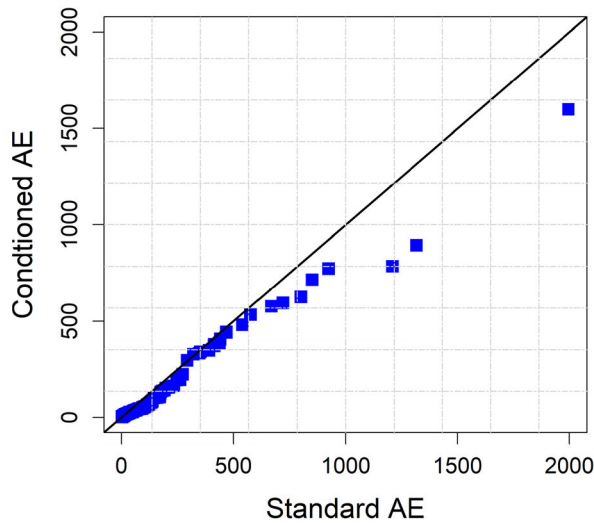


FIGURE 8. Sorted AE of the Standard OPM vs. AE of the Conditioned OPM.

TABLE 6. Confusion matrix for event severity classification for analysis weather data of 14 extratropical events.

Predicted \ True	Low	Moderate	High
Low	4	0	0
Moderate	5	4	0
High	0	0	1

OPM reduces the prediction bias of the Standard OPM. This is because the predictive accuracy of the OPM is sensitive to the representativeness of the training weather events dataset, and the ML models can recognize the severity of the testing event when the training event has similar weather patterns and information.

C. FORECAST PERFORMANCE

This section discusses event severity classification results for 14 analysis and forecast extratropical events and the performance of the Conditioned OPM in predicting them. Tables 6 and 7 show the confusion matrix of event severity classification results using analysis weather data and forecast weather data, respectively, for these 14 events. The classification Accuracy results were 0.64 and 0.57, respectively. The classification F1 Scores for low-, moderate-, and high-severity classes using analysis weather data were 0.62, 0.62, and 1, respectively, while using forecast weather data, they were 0.57, 0.67, and 1, respectively. Significantly, all the low-severity events were classified as low-severity events and all the high-severity events as high-severity events. The classification results using the analysis weather data and the forecast weather data, respectively, were similar for moderate-severity events. The QWD-based event classification for these 14 events using forecast weather data exhibited acceptable performance.

TABLE 7. Confusion matrix for event severity classification for forecast weather data of 14 extratropical events.

Predicted \ True	Low	Moderate	High
Low	4	0	0
Moderate	6	3	0
High	0	0	1

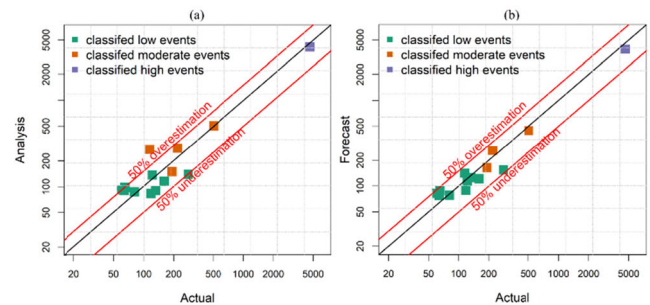


FIGURE 9. Using Conditioned OPM to predict 14 extratropical events using analysis weather data (a) and forecast weather data (b) for comparison.

TABLE 8. Error metrics of conditioned OPM predicting 14 extratropical events with forecast and analysis weather data.

Model	AE Q25	AE Q50	AE Q75	MAPE	CRMSE	R ²	NSE
Analysis	13	36	46	35%	131	0.99	0.99
Forecast	9	26	39	20%	185	0.99	0.97

The scatter plots in Figure 9 show the results of using the Conditioned OPM to predict outages for the 14 extratropical events using analysis (panel a) and forecast weather data (panel b). Table 8 summarizes the error metrics AE Q25, AE Q50, AE Q75, MAPE, and CRMSE of the Conditioned OPM using analysis (panel a in Figure 9), and forecast weather data (panel b), respectively.

In terms of error differences between predictions with forecast and analysis weather data, AE Q25, AE Q50, and AE Q75 increased by 4, 10, and 7 outages, respectively; MAPE increased by 15%; CRMSE decreased by 54 outages; As the high-severity event using analysis and forecast weather data were both predicted very accurately, and this event dominated the error metrics of the conditioned OPM, the R-squared of the models using the analysis and forecast were 0.99; and NSE increased by 0.02. If we calculate the R-squared not including this high-severity event, the R-squared of the models using the analysis and forecast would be 0.7 and 0.87, respectively. The Conditioned OPM using both analysis and forecast weather data showed good performance. Therefore, as evidenced by these results, we could apply the Conditioned OPM using forecast weather data to predict the outages of an upcoming event.

VII. CONCLUSION

This paper describes a technique for improving power distribution grid outage prediction by incorporating storm severity classification of weather-related outage events in an OPM. Storm severity was derived from a classification decision tree based on the QWD method that quantifies the differences in the CDFs of weather variables between the forecasted event and events in the OPM training dataset. Confusion matrix and classification performance statistics showed that the QWD method can accurately categorize events into three severity levels: low, moderate, and high.

The paper compared the Standard OPM performance to the Conditioned OPM performance, facilitated by the QWD-based event classification method, using 100 extratropical events. We showed that the Conditioned OPM could reduce bias significantly by selecting a representative training dataset for each predicted event classified by the QWD-based severity classification method.

This study also investigated the effectiveness of applying the QWD-based event severity classification method and the Conditioned OPM to forecasted weather-related outages in Eversource Energy's Connecticut service territory based on 14 outage events in 2017. The study showed consistent performance of the method and a potential for successful application to outage forecasting.

Despite the overall improved forecast skill of the conditioned OPM, QWD misclassified some moderate and low-severity events affecting the accuracy of the model. To improve this moderate vs. low-severity classification, in future research we will investigate using additional static variables, such as vegetation and land cover, in the calculation of QWD.

This investigation, based on Eversource Energy's Connecticut service territory, could be applied to other service territories to evaluate the scalability of the method. Other future extensions of this study is to apply the severity classification on other storm events (such as convective and snow/ice precipitation events) and evaluate benefits in outage prediction modeling.

APPENDIX

A. KEY WEATHER VARIABLE SELECTION IN QWD STEP DECISIONS

In the step two decision, we calculated the QWD of CDFs between each of the 61 moderate-severity events (X in equation (1)) and each of the 10 high-severity events (Y in equation (1)), as well as the QWD of CDFs between each of the 61 moderate-severity events (X in equation (1)) and each of the remaining 60 moderate-severity events (Y in equation (1)) for the 23 normalized significant weather variables. We designated the QWD of CDFs among the 61 moderate-severity events as blue star points and the QWD of CDFs between each of the 61 moderate-severity events and each of the 10 high-severity events as boxplots. We calculated the average percentage (y-axis in Figure 10) of the blue star points of these 61 moderate-severity events below the 65%,

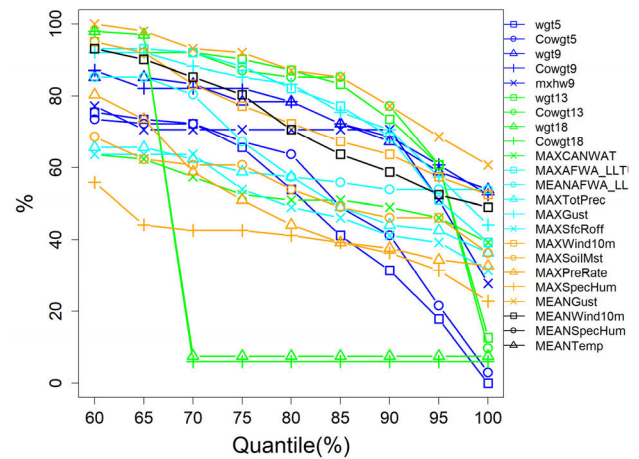


FIGURE 10. Average percentage (y-axis) of blue star points of these 61 moderate-severity events higher than different quantiles of 65%, 70%, 75%, 80%, 85%, 90%, 95%, and 100% (x-axis) of their corresponding boxplots in the step two decision.

70%, 75%, 80%, 85%, 90%, 95%, and 100% quantiles (x-axis in Figure 10) of their corresponding boxplots. From Figure 10, we picked the six key variables (MEANGust, wgt13, wgt18, MAXLLTURB, MAXGust, and MAXWind10m) that exhibited the highest average percentage of blue star points of these 61 moderate-severity events above the 65% quantile of their corresponding boxplots, and we set the QWD values at the 65% quantile of the boxplots as the threshold of the step two decision.

In the step three decision, we calculated the QWD of CDFs between each of the 31 low-severity events (X in equation (1)) and each of the remaining 71 events (Y in equation (1)), as well as the QWD of CDFs between each of the 31 low-severity events (X in equation (1)) and each of the remaining 30 low-severity events (Y in equation (1)) for the 23 normalized significant weather variables. We designated the QWD of CDFs among the 31 low-severity events as blue star points and the QWD of CDFs between each of 31 low-severity events and each of the remaining 71 events as boxplots. We calculated the average percentage (y-axis in Figure 11) of the blue star points of these 31 low-severity events higher than the 65%, 70%, 75%, 80%, 85%, 90%, 95%, and 100% quantiles (x-axis in Figure 11) of their corresponding boxplots. From Figure 11, we picked the five key variables (MAXWind10m, MEANWind10m, wgt9, Cowgt9, and MEANGust) that exhibited the highest average percentage of blue star points of these 10 high-severity events higher than the 65% quantile of their corresponding boxplots, and we set the QWD values at the 65% quantile of the boxplots as the threshold of the step three decision.

In the step four decision, we calculated the QWD of CDFs between each of the 61 moderate-severity events (X in equation (1)) and each of the 31 low-severity events (Y in equation (1)), as well as the QWD of CDFs among the 61 moderate-severity events for the 23 normalized significant weather variables. As in the first decision, we designated the QWD of CDFs among the 61 moderate-severity events as

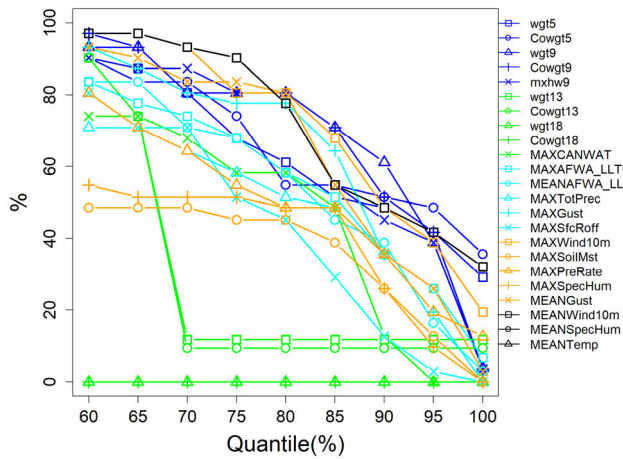


FIGURE 11. Average percentage (y-axis) of blue star points of these 31 low-severity events higher than the 65%, 70%, 75%, 80%, 85%, 90%, 95%, and 100% quantiles (x-axis) of their corresponding boxplots in the step three decision.

blue star points, and the QWD of CDFs between each of the 61 moderate-severity events and each of the remaining 31 low-severity events as boxplots. Using similar methods, we calculated the average percentage of blue star points of these 61 moderate-severity events below different quantiles of their corresponding boxplots, chose QWD values at the 10% quantile of the boxplots as the threshold of the step four decision, and picked six key variables (MEANWind10m, MAXWind10m, mxhw9, MAXLLTURB, MEANGust, and wgt9).

B. CLASSIFICATION METRICS

For performance evaluation of multiclass event classification problems, Accuracy is expressed as the proportion of correct predicted events over all predicted events, which is calculated as follows:

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Number of all predictions}}, \quad (4)$$

For a given class label X (X is low, moderate, or high), terms are defined as follows:

True Positive (TP): Observation is label X and is predicted as label X .

False Positive (FP): Observation is not label X but is predicted as label X .

False Negative (FN): Observation is label X but is not predicted as label X .

True Negative (TN): Observation is not label X and is not predicted as label X .

Precision is expressed as the proportion of events correctly predicted as label X over all events predicted as label X . Precision is calculated as follows:

$$Precision = \frac{TP}{TP + FP}, \quad (5)$$

For a given class label X (X is low, moderate, or high), Recall is the proportion of events correctly predicted as

label X over observed events of label X . Recall is calculated as follows:

$$Recall = \frac{TP}{TP + FN}, \quad (6)$$

Combining metrics of Precision and Recall, the F1 Score is the harmonic mean of Precision and Recall. The F1 Score gives equal weight to Precision and Recall. A maximized F1 Score could create a balanced classification model, and is calculated as follows:

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}, \quad (7)$$

C. MACHINE LEARNING MODELS

1) BAYESIAN ADDITIVE REGRESSION TREE

Bayesian Additive Regression Tree (BART) [45] is derived from Bayesian classification and the regression trees model (CART). We used the R package of “BART,” set 30 trees, and used 10,000 iterations in a backfitting Markov Chain Monte Carlo (MCMC) [46] algorithm to reach convergence and generate the predictions. We averaged 10,000 predictions and got the final output for BART.

2) RANDOM FOREST REGRESSION

Random forest (RF), created by Ho [47] and extended by Breiman [48], is a nonparametric statistical technique that uses bagged versions of decision trees and obtains a lower test error by variance reduction. A higher reduction in variance can be achieved by decreasing the correlation among trees in the ensemble. The forest, therefore, should be sufficiently large to produce reliable predictions for RF.

3) BOOSTED GRADIENT TREE REGRESSION

Boosted gradient tree (BT) is an ensemble technique that builds several small trees called “weak learners” to reduce the loss of function on previous decision trees and generates a “strong learner” to obtain robust predictive models [49]. In this technique, the predictors are not made independently, but sequentially. Specifically, new predictors learn from errors committed by previous ones, with the result that fewer iterations are needed to approach actual predictions. Since the size of regression trees is a regularization parameter that controls the overfitting, an optimal number of decision trees is required in this technique.

4) DECISION TREE REGRESSION

Decision tree regression (DT) [50] is a simple ML model constructed in the form of a tree structure. A decision tree with its root node at the top splits the labeled dataset into branches, which associate the explanatory variables with power outages, and ends when it reaches the leaf node that cannot be partitioned.

5) ERROR METRICS

We used absolute error (AE) to measure the difference between actual (o_i) and predicted (p_i) totals of service

TABLE 9. Storm list of 14 forecast events in Section VI.C.

Date	Actual Outages
2017-05-05	115
2017-05-13	81
2017-05-18	160
2017-06-16	118
2017-07-29	65
2017-08-29	64
2017-09-03	132
2017-09-05	279
2017-09-19	122
2017-10-09	218
2017-10-15	60
2017-10-24	501
2017-10-29	4,626
2017-11-10	192

territory outages from each event (i). The first, second, and third quantiles of the sorted absolute error data were represented by AE Q25, AE Q50, and AE Q75, respectively. AE is calculated as follows:

$$AE = |p_i - o_i|, \quad (8)$$

The first, second, and third quantiles of the sorted absolute percentage error data were represented by APE Q25, APE Q50, and APE Q75, respectively. APE is calculated as follows:

$$APE = \frac{|p_i - o_i|}{o_i}, \quad (9)$$

We used mean absolute percentage error (MAPE) to measure the mean relative error as a percentage. It is calculated as follows:

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{o_i - p_i}{o_i} \right|, \quad (10)$$

We used centered root mean square error (CRMSE) to quantify the random component of the error. It is defined as follows:

$$CRMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(p_i - o_i - \frac{\sum_{i=1}^n (p_i - o_i)}{n} \right)^2}, \quad (11)$$

We used the coefficient of determination R-squared (R^2) to measure the goodness of fit of the various model predictions to the actual outages. It is defined as follows:

$$R^2 = \frac{\frac{1}{n} \sum_{i=1}^n \left(o_i - \frac{\sum_{i=1}^n o_i}{n} \right) \left(p_i - \frac{\sum_{i=1}^n p_i}{n} \right)}{o_i p_i}, \quad (12)$$

We used Nash-Sutcliffe efficiency (NSE), ranging between negative infinite and 1, to determine how well the prediction fit the actual outages. NSE is defined as follows:

$$NSE = 1 - \frac{\sum_{i=1}^n (o_i - p_i)^2}{\sum_{i=1}^n \left(o_i - \frac{\sum_{i=1}^n o_i}{n} \right)^2}, \quad (13)$$

D. STORM LIST OF FORECAST EVENTS

See Table. 9.

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E. N. Anagnostou holds stock in ACW Analytics. The authors have full access to all of the data in this study, and they take complete responsibility for the integrity of the data and the accuracy of the data analysis.

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FEIFEI YANG received the B.S. degree in environmental engineering from the Anhui University of Technology and the M.S. degree in environmental engineering from the Dalian University of Technology. She is currently pursuing the master's degree in computer science and engineering with the University of Connecticut and the Ph.D. degree with the Department of Civil and Environmental Engineering. She worked as an Environmental Engineer and a Researcher at China Sinopec, from 2011 to 2016. Her major Ph.D. research focuses on the prediction of weather-related power outages in terms of machine learning and statistics.



PETER WATSON received the B.A. degree in environmental studies from the University of Chicago and the M.S. degree in environmental engineering from the University of Connecticut, where he is currently pursuing the Ph.D. degree in environmental engineering. His researches focus on machine learning-based impact models, their spatial structure, and their use as decision support tools.



MARIKA KOUKOULA received the B.A. degree in geography from the Harokopio University of Athens and the M.S. degree in environmental physics and meteorology from the University of Athens. She is currently pursuing the Ph.D. degree in environmental engineering with the University of Connecticut. Her researches focus on land surface—atmosphere interactions, extreme events, and atmospheric modeling.



EMMANOUIL N. ANAGNOSTOU is currently a Board of Trustees Distinguished Professor of civil and environmental engineering and the Eversource Energy Endowed Chair in environmental engineering with the University of Connecticut. He is also the Founding Director of the Eversource Energy Center and the Applied Research Director of the Connecticut Institute for Resilience and Climate Adaptation. He currently coordinates a National Science Foundation-Partnership for International Research and Education multiinstitutional food security project in Ethiopia. In his capacity as the Eversource Energy Center Director and one of CIRCA's Applied Research Directors, he is leading power grid resilience and inland, and coastal flood vulnerability studies. His research expertise is in remote sensing applications in water resources.

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